AlloyDB for PostgreSQL - Michaël Bettan

Definition: Fully-managed, PostgreSQL-compatible database service designed for demanding enterprise database workloads. Combines a Google-built database engine with a disaggregated, log-structured storage layer. Optimized for high-throughput OLTP, real-time analytics (HTAP), and AI-driven applications.	**Availability**: Regional (Multi-zone HA) Omni (Run anywhere)	**SLA**: 99.99%

Use Cases:

- ***Tier-1 OLTP**: mission-critical operational database applications.*
- ***HTAP & Real-time Analytics**: hybrid transactional and analytical processing.*
- ***Legacy Modernization**: modernizing legacy systems, including migrations from Oracle.*
- ***Generative AI & Vector Search**: build intelligent applications leveraging integrated AlloyDB AI.*
- ***High-throughput Scaling**: workloads exceeding standard Cloud SQL performance limits.*

Billing:

- **Compute**: vCPU & Memory per hour for Primary and Read pool nodes (HA doubles primary cost)
- **Storage**: Data storage per GiB/month (pay-for-what-you-use, no provisioning required)
- **Network Egress**: Data transferred out
- **Backup storage**: Snapshot and PITR log storage.
- No I/O charges
- PITR logs are free for the first 7 days

Architecture & Storage Model

- **Decoupled Compute & Storage**: *Compute nodes and storage scale independently. Storage is a highly available distributed service shared across the cluster.*
- **Log-Structured Storage Layer**: *Compute nodes do not write data blocks to storage; they only write Write-Ahead Logs (WAL). The storage layer autonomously applies WALs to blocks, drastically reducing I/O bottlenecks and vacuuming overhead.*
- **Primary Instance (Compute Layer)**: *A single read/write node. If configured for HA, it includes an active/standby pair across zones. Max capacity is 288 vCPUs and 2,232 GiB RAM per instance.*
- **Read Pool Instances (Compute Layer)**: *Up to 20 nodes for read scaling, no replication lag (all nodes share the same distributed storage layer, writes on primary are immediately visible to all read pool nodes)*
- **Storage Layer**: *Replicated automatically across 3 zones in a region for durability. Auto-scales seamlessly up to 128 TiB per cluster.*
- **Cache Hierarchy**: *Multi-layer caching (Buffer Cache, Ultra-fast Local SSDs) significantly accelerates data access compared to standard PostgreSQL.*

Key Capabilities

- **100% PostgreSQL Compatibility**: *Works out-of-the-box with standard PostgreSQL drivers, ORMs, and extensions.*
  - *Supports PostgreSQL major versions. Minor version patches are applied automatically,*
- **Performance**: *Up to 4x faster for transactional workloads (OLTP) and up to 100x faster for analytical queries (OLAP) than standard open-source PostgreSQL.*
- **Auto-scaling storage**: *Storage capacity grows automatically as data increases without requiring downtime or resizing operations.*
- **Connection management**: *An instance supports up to 1,000 concurrent connections by default, but AlloyDB has a native Managed Connection Pooler. Max scales to 240,000 concurrent connections. Standard*
  - *PostgreSQL port: 5432*
  - *Managed Connection Pooler port: 6432*

HTAP & Columnar Engine

- **Built-in Columnar Engine**: *Keeps a column-based representation of data in memory, specifically optimized for analytical queries (aggregations, scans).*
- **Auto-columnarization (Machine Learning)**: *ML algorithms observe workload patterns and automatically decide which tables/columns to load into the columnar engine.*
- **Real-Time Insights**: *Enables running complex BI reporting directly on the operational database without needing ETL pipelines to a separate data warehouse.*

AlloyDB AI & Machine Learning

- **Vertex AI Integration**: *Integrates seamlessly with Vertex AI to call ML models directly from SQL (e.g., using google_ml.predict_row()) for inference and generating embeddings.*
- **Vector Search**: *Leverages Google's ScaNN index (the same technology powering Google Search/YouTube) to deliver up to 10x faster index creation, 4x faster pure vector search, and 10x faster filtered vector searches than the standard pgvector HNSW index.*
- **Generative AI & RAG**: *Enables building scalable Generative AI and RAG applications directly on transactional data.*
- **Auto Vector Embeddings**: *Automatically synchronizes embeddings with transactional data. Supports incremental refresh for updated rows and a bulk backfill mode 130x faster than standard row-by-row processing.*
- **google_ml.generate_content**: *Invokes Large Language Models (e.g., gemini-2.0-flash-001) directly from SQL for RAG workflows, leveraging the google_ml_integration extension.*

AlloyDB Omni

- **Downloadable Edition**: *Containerized version of AlloyDB designed to run anywhere.*
- **Deployment Flexibility**: *Run on-premises, on developer laptops (via Docker/VMs), at the edge, or on competing clouds (AWS, Azure).*

- **Core Engine Optimizations**: *Brings the exact same core engine optimizations (columnar engine, vector search, ScaNN) to self-managed infrastructure.*

Migration & Integrations

- **Database Migration Service (DMS)**: *Native integration for secure, minimal-downtime migrations from legacy databases into AlloyDB with Gemini-assisted schema conversion for heterogeneous migrations (e.g., SQL Server → PostgreSQL)*
    - *Amazon RDS, Amazon Aurora, MySQL, Microsoft SQL Server, Oracle, Self-managed Postgres*
- **Datastream**: *Change Data Capture (CDC) to stream data from AlloyDB into BigQuery for long-term enterprise data warehousing.*

Security & Networking

- **Private Access**: *Connect securely via VPC peering to access instances through private IPs without traversing the public internet.*
- **Encryption**: *Data is encrypted in transit and at rest, supporting Customer-Managed Encryption Keys (CMEK).*
- **IAM Authentication**: *Centralized access control leveraging Google Cloud IAM.*
- **VPC Service Controls**: *Establish secure perimeters around database resources.*
- **Audit Logging**: *Detailed Cloud Audit Logs for database access and administrative actions.*
- *Supports **Private Service Connect (PSC)***

High Availability & Backup section

- **Rapid HA Failover**: *Automatic failover completes in < 60 seconds, independent of database size.*
- **Disaster Recovery**: *Cross-region replication available for robust DR strategies.*
    - ***Planned switchover** → zero RPO for scheduled maintenance.*
    - ***Unplanned regional outage** → minimal, non-zero RPO (dependent on replication lag at moment of failure, but 25x lower lag than standard PostgreSQL)*
- **Continuous Backup & PITR**: *Point-in-time recovery retained for 14 days by default (adjustable from 1–35 days).*
- **Intra-region Recovery**: *Provides zero RPO for point-in-time recovery scenarios within a region.*